

EXPERIMENT 02B

Medical GPT Training Report using Custom Biomedical BPE Tokenizer

Medical Domain Corpus | ~5 Million Tokens | Custom Biomedical BPE Tokenizer

Dataset	Hardware	Date	Epochs
Medical Corpus (~5M tokens)	Google Colab T4	May 2026	5

1. Objective

Experiment 02B evaluates a GPT-style decoder-only Transformer trained on the same biomedical corpus and identical architecture used in Experiment 02A while replacing the GPT-2 tokenizer with a custom biomedical BPE tokenizer.

The primary objective was to measure the effect of domain-specific tokenization on language model training efficiency, validation performance, and perplexity while keeping all other training variables constant.

This experiment serves as a controlled tokenizer comparison study in which the tokenizer is the only modified component.

2. Model Configuration

The model architecture remained unchanged from Experiment 02A to ensure a fair tokenizer evaluation.

Component	Configuration	Parameters (Approx.)
Token Embedding	52,000 × 512	26.6 M
Position Embedding	256 × 512	0.13 M
Transformer Layers	6 Layers d_model=512 heads=8	~18–20 M
LM Head	52,000 × 512 Untied	26.6 M
Total	Decoder-only Transformer	~70 M

The custom tokenizer vocabulary consisted of approximately 52,000 learned tokens generated from a biomedical corpus using Byte Pair Encoding (BPE).

3. Hyperparameter Configuration

Parameter	Value
Vocabulary Size	~52,000
Context Length	256

Parameter	Value
d_model	512
Attention Heads	8
Transformer Layers	6
Dropout	0.1
Learning Rate	3e-4
Minimum Learning Rate	1e-5
Gradient Accumulation	8 Steps
Mixed Precision (AMP)	Enabled

All hyperparameters remained identical to Experiment 02A.

4. Training Setup

Setting	Value
Optimizer	AdamW
Weight Decay	0.1
Gradient Clipping	1.0
Learning Rate Schedule	Cosine Decay
Sliding Window Stride	64 Tokens
Checkpoint Strategy	latest.pt + best.pt + epoch_N.pt
Flash Attention	Not Used

The same training infrastructure and optimization pipeline used in Experiment 02A were retained to isolate tokenizer effects.

5. Training Results

Epoch	Train Loss	Validation Loss	Perplexity
1	3.4282	2.7003	14.8835
2	2.5097	2.3063	10.0374
3	2.2175	2.1228	8.3544
4	2.0608	2.0381	7.6764
5	1.9890	2.0155	7.5042

The model exhibited stable convergence throughout all training epochs with continuous improvement in validation loss and perplexity.

6. Analysis

Validation loss decreased consistently from 2.7003 to 2.0155 across five epochs, while perplexity declined from 14.8835 to 7.5042.

Compared to Experiment 02A, convergence occurred substantially faster and achieved significantly lower validation loss throughout training. The reduction in perplexity indicates that the model was able to represent biomedical language patterns more effectively when trained using the custom tokenizer.

Training and validation losses remained closely aligned during optimization, indicating stable generalization without severe overfitting.

7. Root Cause Analysis

Factor	Status	Observation
Domain-Specific Tokenization	Improved	Biomedical terminology represented more efficiently
Vocabulary Construction	Improved	Frequent medical subwords learned directly
Token Sequence Efficiency	Improved	Reduced fragmentation of medical terminology
Training Stability	Improved	Faster convergence observed throughout training
Generalization	Improved	Lower validation loss and perplexity achieved

The custom tokenizer reduced the number of fragmented subword sequences required to represent specialized medical vocabulary. This allowed the model to learn domain-specific linguistic patterns more efficiently.

8. Comparison with Experiment 02A

Metric	Experiment 02A	Experiment 02B
Dataset	Same	Same
Architecture	Same	Same
Parameters	~70 M	~70 M
Tokenizer	GPT-2 BPE	Custom Biomedical BPE
Final Train Loss	3.1951	1.9890
Final Validation Loss	4.0208	2.0155
Final Perplexity	55.7466	7.5042
Validation Trend	Decreasing	Decreasing
Generalization	Stable	Stronger

The custom biomedical tokenizer produced significantly lower validation loss and perplexity despite identical training conditions, demonstrating the effectiveness of domain-specific tokenization for medical language modeling.

9. Conclusion

Experiment 02B successfully demonstrated the benefits of domain-specific tokenization for biomedical language modeling. By replacing the GPT-2 tokenizer with a custom biomedical BPE tokenizer while maintaining an identical training configuration, the experiment isolated tokenizer performance as the primary experimental variable.

The custom tokenizer achieved substantially lower validation loss and perplexity than Experiment 02A, indicating more efficient representation of medical terminology and improved language modeling capability. The final validation loss of 2.0155 and perplexity of 7.5042 significantly outperformed the GPT-2 TikToken baseline established in Experiment 02A, demonstrating the effectiveness of domain-specific tokenization for biomedical language modeling.

Experiment 02B validates the effectiveness of the custom tokenizer design and establishes a strong foundation for future large-scale training experiments, including Experiment 03A and subsequent tokenizer optimization studies.